

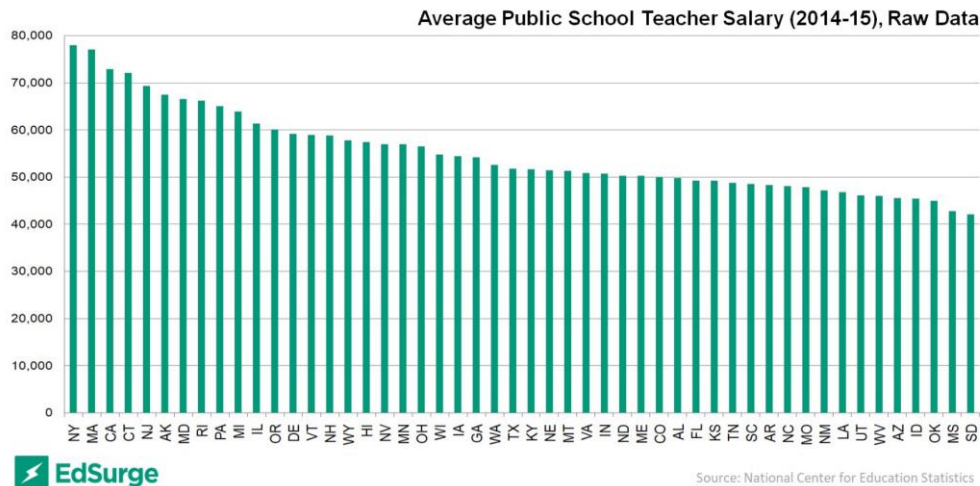
Describing and Comparing Distributions

In the world of statistics, it is not enough to just report the graph or just report the summary statistics.

Why?

The graph alone might not give us all the information we need to make a valid conclusion.

Ex. Do teachers in New York get paid too much? The graph below shows the average teacher salary for each state (from 2014/15).



This graph shows that teachers in New York state make almost double than teachers in South Dakota. Is this a valid conclusion or is the graph not telling the whole story?

The summary statistics might not give us all the information we need to make a valid conclusion.

Ex. The mean teacher salary in the state of California (from 2019/20 data) is \$84,531. This lets us know that teachers are making above the median income for everyone in California, which is \$78,672. Is this a valid conclusion or are the summary statistics not telling the whole story?

Being the great statistics students you are, you must report both a visual display of the data as well as detailed summary statistics when asked to “describe the distribution”:

First: Create your graph (if one is not given)

- Is your data categorical? Use a bar graph! (With two variables next unit, we might also use a segmented bar graph or a mosaic plot)
- Is your data quantitative? (Two variables will require a scatterplot!)
 - Discrete? Use a dotplot, stemplot, or boxplot.
 - Continuous? Use a histogram or a boxplot.

Cumulative relative frequency graphs will usually take too long to create on the AP exam, but remember if you are interested in percentile positions, that you can create these graphs as well!

Second: Summarize your findings

We’ve discussed the acronym “SOCS” as a way to remember the 4 things to comment on once your graph is created.

- Shape – Skewed, Symmetric, Bimodal, etc.
 - Be as specific as possible and combine shape terms if you can.
- Outliers – an individual observation that falls outside the overall pattern of the graph.
 - You will just have to comment on if there are any visual outliers. You do not have to do the outlier test unless they instruct you to.
- Center – Mean and Median
 - Use the mean, unless the distribution is skewed or has outliers.
 - Unless asked for otherwise, a verbal description of the center will do.
- Spread – Range, Standard Deviation, IQR
 - If you describe the center with the mean, comment on the standard deviation.
 - If you describe the center with the median, comment on the range or IQR.

When the problem asks you to “compare the distributions”, you want to follow the above steps, but you want to make sure that you are using comparison words to compare the distributions.

Do not do...	What you should do...
The median teacher salary in California is \$61,595 and the median teacher salary in Arizona is \$47,606.	The median teacher salary of \$61,595 in California is higher than the median teacher salary of \$47,606 in Arizona.

Example: The following data are for two popular songs on the Billboard Top 100 (occurring during different decades). The length of each word in the song was recorded and below shows the number of words with the corresponding number of letters.

Sweet Child O'Mine by Guns and Roses

Length of Word	1	2	3	4	5	6	7	8	9	10
Number of Words	9	109	42	51	47	2	6	2	1	1

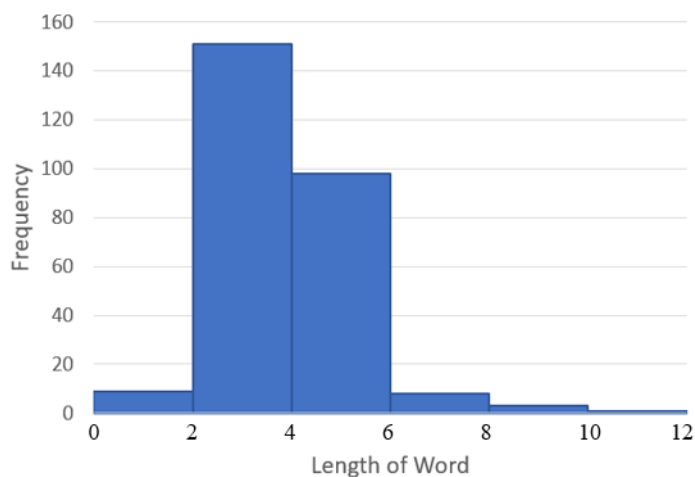
Butter by BTS

Length of Word	1	2	3	4	5	6	7	8	9	10
Number of Words	13	77	91	137	36	26	6	3	2	1

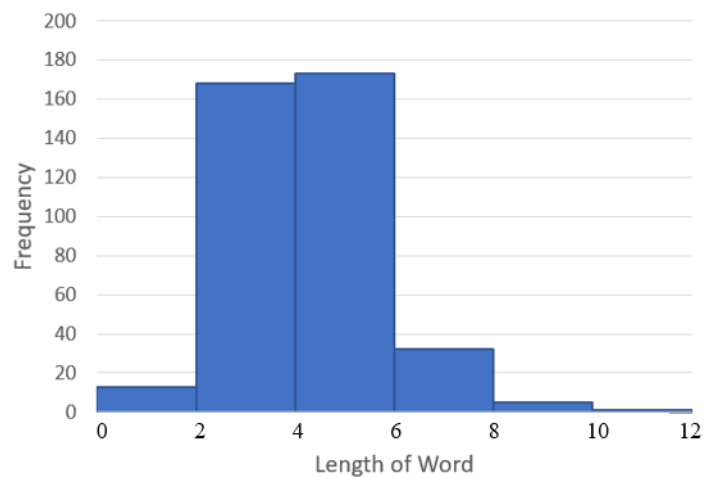
(a) Determine the five number summary for each data set.

(b) Below are the graphs of the two distributions. Write a few sentences comparing the distributions.

Sweet Child O'Mine



Butter



(c) There are two “rules” we use to mathematically determine outliers. Method A is the $1.5 \times \text{IQR}$ rule and Method B is the two standard deviation rule.

(i) Using method A, determine the outliers that are present in the Sweet Child O’Mine distribution. Justify your answer.

(ii) The mean number of letters in the Butter distribution is 3.62 and the standard deviation is 1.42. Using method B, determine the outliers that are present in the Butter distribution. Justify your answer.

(d) Explain why method A is better for determining outliers than method B in these distributions.